

# Alcohol intake and mortality among women with invasive breast cancer

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Background: Alcohol intake has consistently been associated with increased breast cancer incidence in epidemiological studies. However, the relation between alcohol and survival after breast cancer diagnosis is less clear. Methods: We investigated whether alcohol intake was associated with survival among 3146 women diagnosed with invasive breast cancer in the Swedish Mammography Cohort. Alcohol consumption was estimated using a food frequency questionnaire. Cox proportional hazard models were used to calculate hazard ratios (HRs) and 95% confidence intervals (95% CIs). Results: From 1987 to 2008 there were 385 breast cancer-specific deaths and 860 total deaths. No significant association was observed between alcohol intake and breast cancer-specific survival. Women who consumed 10 g per day (corresponding to approximately 0.75 to 1 drinks) or more of alcohol had an adjusted HR (95% CI) of breast cancer-specific death of 1.36 (0.82–2.26;  $p_{\text{trend}}=0.47$ ) compared with non-drinkers. A significant inverse association was observed between alcohol and non-breast cancer deaths. Those who consumed 3.4–9.9 g per day of alcohol had a 33% lower risk of death compared with non-drinkers (95% CI 0.50–0.90;  $p_{\text{trend}}=0.04$ ). Conclusion: Our findings suggest that alcohol intake up to approximately one small drink per day does not negatively impact breast cancer-specific survival and a half drink per day is associated with a decreased risk of mortality from other causes.